Frames and matrix designs

Vanishing injectivity probability at 4d minus 5 complex phase measurements

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: The asymptotic typical-injectivity question needs quantitative algebraic and probabilistic control beyond positive failure probability; it informs phase-retrieval measurement design.

For every integer $d \geq 2$, draw $A_d \in \mathbb{C}^{(4d-5)\times d}$ with independent standard complex Gaussian entries: the real and imaginary parts are independent $N(0, 1/2)$. Let p_d be the probability that

$$|A_d x| = |A_d y| \quad \Longrightarrow \quad y = e^{i\theta} x \text{ for some } \theta \in \mathbb{R}$$

holds for all $x, y \in \mathbb{C}^d$, where absolute value is componentwise. Equivalently, p_d is the probability that the induced phase-retrieval map on vectors modulo global phase is injective.

Is

$$\lim_{d\to\infty} p_d = 0?$$

This is part (b) of Vinzant's conjecture as restated in Randomstrasse 101. It distinguishes injective exceptional measurement systems from typical matrices at a row count just below $4d-4$. Injectivity is the basic identifiability requirement before conditioning and stable numerical inversion can be addressed.

References

1. A. S. Bandeira et al., *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 19(b). Paper.
2. C. Vinzant, *A small frame and a certificate of its injectivity*, SAMPTA 2015, arXiv:1502.04656. The explicit 11-vector frame in $\mathbb{C}^4$ demonstrates why the below-$4d-4$ regime cannot simply be dismissed. Paper.
3. Z. Li, *On Injectivity of Phase Retrieval*, arXiv:2606.17922 (2026). The abstract and main result establish the strict inequality $p_d < 1$, explicitly identifying this as the first part of Vinzant's conjecture. Paper.

Status check — 2026-09-10

Searched "Vinzant conjecture injectivity probability limit", "phase retrieval 4M-5 2026", and checked the latest June 2026 paper abstract/version. Its positive probability of noninjectivity does not supply an asymptotic lower bound tending to one. The already proved assertion $p_d < 1$ is deliberately excluded from this problem.

Audit update (2026-09-10): Rechecked Li's June 2026 abstract and searched for the asymptotic part of Vinzant's conjecture. The reported strict inequality $p_d < 1$ does not force $p_d \to 0$; this entry contains only the latter target. This is a bounded literature check, not a proof that no solution exists.